

AR 526: Multimodal Learning

Course Code	AR 526
Course Name	Multimodal Learning
L-T-P-C	3-0-0-3
Prerequisites	IC 114: Linear Algebra/ Introductory course in machine-learning
Intended for	UG, PG and PhD
Mutual Exclusion	None
Approval	48th Senate (Mar 2026), in the B.Tech General Engineering with Mechatronics & AI curriculum; 64th BoA (May 2026)

Preamble

Recent advances in artificial intelligence and robotics have underscored the importance of multimodal learning for enabling machines to understand, reason, and act using diverse sources of information such as text, vision, audio, and sensor data. In robotic systems, the ability to fuse and align multiple modalities is critical for perception, navigation, and interaction in complex real-world environments. This course offers a comprehensive introduction to multimodal learning, with a focus on its applications in natural language processing, computer vision, audio analysis, and robotics. Through hands-on assignments and project work, learners will acquire the skills to build multimodal AI systems capable of performing language-guided navigation, audio-visual perception, sensor fusion, and intelligent decision-making in robotic platforms.

Course Modules

- **Foundations of Multimodal Learning:** Definition and motivation; Key modalities: text, image, audio, video, sensor data; Key Challenges: Alignment, fusion, heterogeneity, missing modalities; Classical vs deep multimodal learning. (6 Hours)
- **Unimodal Representations:** Text: Word2Vec, BERT, LLM embeddings; Vision: CNN, ResNet, ViT; Audio: MFCC, log-mel spectrograms, Wav2Vec; Robotics: Sensor (LiDAR, IMU) embeddings, proprioception; Feature normalization, dimensionality issues; Contrastive pre-training and feature alignment (CLIP, ALIGN). (8 Hours)
- **Multimodal Fusion Techniques:** Early, late, and hybrid fusion; Feature-level vs decision-level fusion; Multimodal alignment; Multimodal transformers: ViLT, LXMERT, and ALBEF; Case studies in multimodal classification & grounding (5 Hours)
- **Cross-Modal Generation and Translation:** Vision-to-text (image captioning, VQA, video summarization, BLIP/BLIP2, Flamingo, LLaVA); Text-to-vision (image synthesis with diffusion models (DALL·E) and GANs); Speech-to-text and text-to-speech translation; Robotics: Language-conditioned behaviour generation; Cross modality loss function analysis; Hands-on: Implementing image captioning using pre-trained BLIP model. (6 Hours)
- **Large Multimodal Models (LMMs):** LLaVA, GPT-4V, Gemini, Kosmos-2, and MM-ReAct architectures; Visual Instruction Tuning (VIT) and Multimodal Chain-of-Thought; Tool-augmented and grounded multimodal agents; Hands-on: Building a lightweight multimodal chatbot (Vision + Text) (9 Hours)
- **Evaluation, Explainability, and Applications:** Metrics: accuracy, retrieval recall, alignment, cross-modal entropy; Explainability in multimodal systems (Grad-CAM, attention visualization); Applications: Robotics, Healthcare, surveillance, AR/VR, and forensics; Ethical considerations: bias, misinformation, and data privacy. (6 Hours)

Laboratory Modules

NA

Text Books

1. Sebe, N., Snoek, C., & Tuytelaars, T. (2024). *Multimodal Machine Learning: Techniques and Applications*. Cambridge University Press.
2. Ding, S., Ong, D. C., & Morency, L.-P. (2022). *Foundations of Multimodal Deep Learning*. Morgan & Claypool Publishers, Synthesis Lectures on Artificial Intelligence and Machine Learning.

References

1. For detailed references, please refer to research papers published in top-tier conferences such as CVPR, ICRA, IROS, ECCV, ICCV, NeurIPS, ICLR.
2. During the lecture, I will provide additional research publications for further reference.